

Q1 – Apache Web Server and Web Application Assessment

Q1 - Web Server Assessment

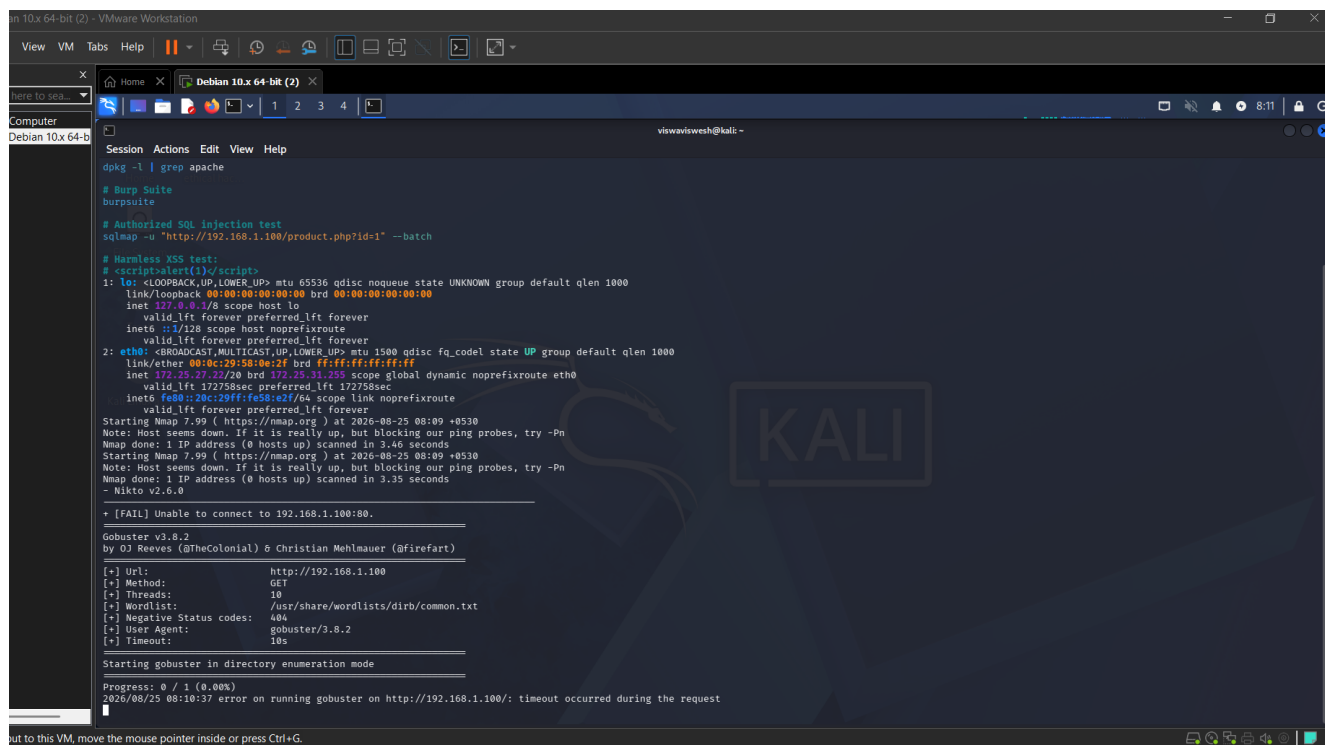
```
ip a
nmap -sV 192.168.1.100
nmap -sV -p- 192.168.1.100
nikto -h http://192.168.1.100
gobuster dir -u http://192.168.1.100 -w /usr/share/wordlists/dirb/common.txt
curl -I http://192.168.1.100
nmap --script http-enum,http-methods -p 80,443 192.168.1.100
apache2 -v
dpkg -l | grep apache
```

burpsuite

```
sqlmap -u "http://192.168.1.100/product.php?id=1" --batch
```

XSS test:

```
# <script>alert(1)</script>
```



```
dpkg -l | grep apache
# Burp Suite
burpsuite
# Authorized SQL injection test
sqlmap -u "http://192.168.1.100/product.php?id=1" --batch
# Harmless XSS test:
# <script>alert(1)</script>
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:58:0a:2f brd ff:ff:ff:ff:ff:ff
    inet 172.25.27.22/20 brd 172.25.31.255 scope global dynamic noprefixroute eth0
        valid_lft 172758sec preferred_lft 172758sec
    inet6 fe80::20c:29ff:fe58:a27f/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-25 08:09 +0530
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.46 seconds
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-25 08:09 +0530
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.35 seconds
- Nikto v2.6.0
+ [FAIL] Unable to connect to 192.168.1.100:80.
Gobuster v3.8.2
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)
[*] Url: http://192.168.1.100
[*] Method: GET
[*] Threads: 10
[*] Wordlist: /usr/share/wordlists/dirb/common.txt
[*] Negative Status codes: 404
[*] User Agent: gobuster/3.8.2
[*] Timeout: 10s
Starting gobuster in directory enumeration mode
Progress: 0 / 1 (0.00%)
2026/08/25 08:10:37 error on running gobuster on http://192.168.1.100/: timeout occurred during the request
```

Q2 – SQL Injection and XSS Assessment

Q2 - SQL Injection and XSS Assessment

```
ip a
nmap -sV 192.168.1.100
nikto -h http://192.168.1.100
```

burpsuite

```

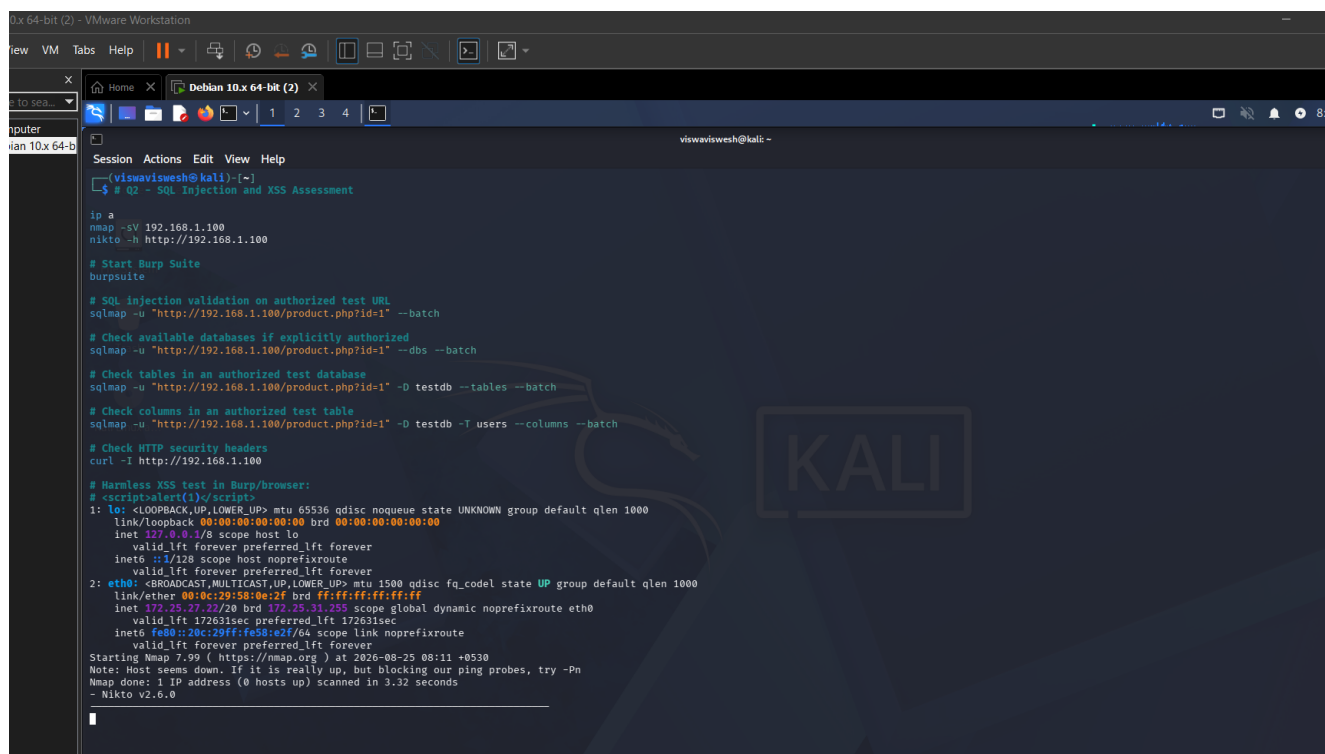
sqlmap -u "http://192.168.1.100/product.php?id=1" --batch
sqlmap -u "http://192.168.1.100/product.php?id=1" --dbs --batch
sqlmap -u "http://192.168.1.100/product.php?id=1" -D testdb --tables --batch
sqlmap -u "http://192.168.1.100/product.php?id=1" -D testdb -T users --columns --batch

```

```
curl -I http://192.168.1.100
```

```
# XSS test:
```

```
# <script>alert(1)</script>
```



```

viswaviswesh@kali: ~
$ # 02 - SQL Injection and XSS Assessment

ip a
nmap -sV 192.168.1.100
nikto -h http://192.168.1.100

# Start Burp Suite
burpsuite

# SQL injection validation on authorized test URL
sqlmap -u "http://192.168.1.100/product.php?id=1" --batch

# Check available databases if explicitly authorized
sqlmap -u "http://192.168.1.100/product.php?id=1" --dbs --batch

# Check tables in an authorized test database
sqlmap -u "http://192.168.1.100/product.php?id=1" -D testdb --tables --batch

# Check columns in an authorized test table
sqlmap -u "http://192.168.1.100/product.php?id=1" -D testdb -T users --columns --batch

# Check HTTP security headers
curl -I http://192.168.1.100

# Harmless XSS test in Burp/browser:
# <script>alert(1)</script>
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:58:0e:2f brd ff:ff:ff:ff:ff:ff
    inet 172.25.27.22/20 brd 172.25.31.255 scope global dynamic noprefixroute eth0
        valid_lft 172631sec preferred_lft 172631sec
    inet6 fe80::20c:29ff:fe58:e27/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-25 08:11 +0530
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.32 seconds
- Nikto v2.6.0

```

Q3 – Evil Twin / Wireless Security Analysis

```
# Q3 - Wireless Security Analysis
```

```
# Use only an authorized lab network
```

```
iwconfig
```

```
ip link
```

```
sudo airmon-ng start wlan0
```

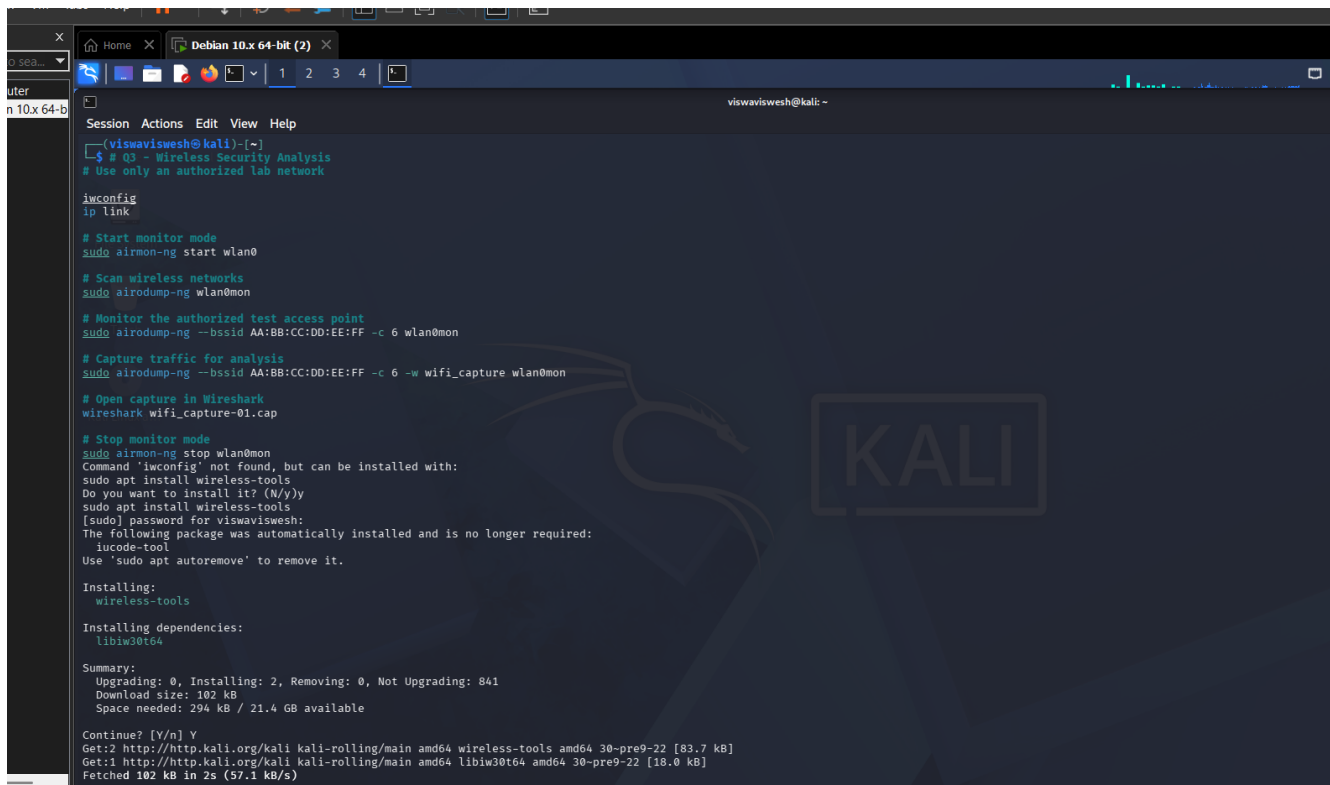
```
sudo airodump-ng wlan0mon
```

```
sudo airodump-ng --bssid AA:BB:CC:DD:EE:FF -c 6 wlan0mon
```

```
sudo airodump-ng --bssid AA:BB:CC:DD:EE:FF -c 6 -w wifi_capture wlan0mon
```

```
wireshark wifi_capture-01.cap
```

```
sudo airmon-ng stop wlan0mon
```



```
(viswaviswesh@kali)~$  
$ # Q3 - Wireless Security Analysis  
# Use only an authorized lab network  
  
iwconfig  
ip link  
  
# Start monitor mode  
sudo airmon-ng start wlan0  
  
# Scan wireless networks  
sudo airodump-ng wlan0mon  
  
# Monitor the authorized test access point  
sudo airodump-ng --bssid AA:BB:CC:DD:EE:FF -c 6 wlan0mon  
  
# Capture traffic for analysis  
sudo airodump-ng --bssid AA:BB:CC:DD:EE:FF -c 6 -w wifi_capture wlan0mon  
  
# Open capture in Wireshark  
wireshark wifi_capture-01.cap  
  
# Stop monitor mode  
sudo airmon-ng stop wlan0mon  
Command 'iwconfig' not found, but can be installed with:  
sudo apt install wireless-tools  
Do you want to install it? (N/y)y  
[sudo] password for viswaviswesh:  
The following package was automatically installed and is no longer required:  
  iucode-tool  
Use 'sudo apt autoremove' to remove it.  
  
Installing:  
  wireless-tools  
  
Installing dependencies:  
  libiw30t64  
  
Summary:  
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 841  
  Download size: 102 kB  
  Space needed: 294 kB / 21.4 GB available  
  
Continue? [Y/n] Y  
Get:2 http://http.kali.org/kali kali-rolling/main amd64 wireless-tools amd64 30-pre9-22 [83.7 kB]  
Get:1 http://http.kali.org/kali kali-rolling/main amd64 libiw30t64 amd64 30-pre9-22 [18.0 kB]  
Fetched 102 kB in 2s (57.1 kB/s)
```

Q4 – Wireless to Web Application Penetration Test

Q4 - Authorized Penetration Test

```
ip addr  
ip route
```

```
nmap -sn 192.168.1.0/24  
nmap -sV -p 80,443 192.168.1.100
```

```
nikto -h http://192.168.1.100  
gobuster dir -u http://192.168.1.100 -w /usr/share/wordlists/dirb/common.txt
```

burpsuite

```
nmap --script http-methods -p 80,443 192.168.1.100
```

```
sqlmap -u "http://192.168.1.100/employee.php?id=1" --batch
```

```
curl -I http://192.168.1.100
```

XSS test:

```
# <script>alert(1)</script>
```

```

(2) - VMware Workstation
M  Tabs  Help  [Icons]
x  Home  Debian 10.x 64-bit (2)  [Icons]
64-bit  viswaviswesh@kali: - 8:13

Session  Actions  Edit  View  Help
(vismaviswesh@kali):~$ # Q4 - Authorized Penetration Test

ip addr
ip route

# Discover hosts
nmap -sn 192.168.1.0/24

# Scan authorized web server
nmap -sV -p 80,443 192.168.1.100

# Web vulnerability scan
nikto -h http://192.168.1.100

# Directory enumeration
gobuster dir -u http://192.168.1.100 -w /usr/share/wordlists/dirb/common.txt

# Start Burp Suite
burpsuite

# Check HTTP methods
nmap --script http-methods -p 80,443 192.168.1.100

# Authorized SQL injection validation
sqlmap -u "http://192.168.1.100/employee.php?id=1" --batch

# Check response headers and cookies
curl -I http://192.168.1.100

# Harmless XSS test in Burp/Browser:
# script=alert(3)/script=
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 08:0c:29:38:0a:2f brd ff:ff:ff:ff:ff:ff
   inet 172.25.27.22/20 brd 172.25.31.255 scope global dynamic noprefixroute eth0
       valid_lft 172538sec preferred_lft 172538sec
   inet6 fe80::20c:29ff:fe58:e2f/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
default via 172.25.16.1 dev eth0 proto dhcp src 172.25.27.22 metric 100
172.25.16.0/20 dev eth0 proto kernel scope link src 172.25.27.22 metric 100
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-25 08:13 +0530

```

Q5 – Wireless Deauthentication Attack Analysis

Q5 - Wireless Deauthentication Analysis

Use only an isolated/authorized lab network

iwconfig

sudo airmon-ng start wlan0

sudo airodump-ng wlan0mon

sudo airodump-ng --bssid AA:BB:CC:DD:EE:FF -c 6 -w deauth_test wlan0mon

wireshark deauth_test-01.cap

Wireshark filter:

wlan.fc.type_subtype == 0x0c

Disassociation filter:

wlan.fc.type_subtype == 0x0a

sudo airmon-ng stop wlan0mon

```
new VM Tabs Help
Debian 10.x 64-bit (2)
viswaviswesh@kali: -
Session Actions Edit View Help
(viswaviswesh@kali)-[~]
$ $ Q5 - Wireless Deauthentication Analysis
# Use only an isolated/authorized lab network

iwconfig

# Start monitor mode
sudo airmon-ng start wlan0

# Scan wireless networks
sudo airodump-ng wlan0mon

# Capture traffic from authorized test AP
sudo airodump-ng --bssid AA:BB:CC:DD:EE:FF -c 6 -w deauth_test wlan0mon

# Open capture in Wireshark
wireshark deauth_test-01.cap

# Wireshark filter for deauthentication frames:
# wlan.fc.type_subtype == 0x0c

# Wireshark filter for disassociation frames:
# wlan.fc.type_subtype == 0x0a

# Stop monitor mode
sudo airmon-ng stop wlan0mon
lo      no wireless extensions.
eth0    no wireless extensions.

[sudo] password for viswaviswesh:

Found 1 processes that could cause trouble.
Kill them using 'airmon-ng check kill' before putting
the card in monitor mode, they will interfere by changing channels
and sometimes putting the interface back in managed mode

PID Name
886 NetworkManager

Requested device "wlan0" does not exist.
Run /usr/sbin/airmon-ng without any arguments to see available interfaces
Interface wlan0mon:
ioctl(SIOCGIFINDEX) failed: No such device
Failed initializing wireless card(s): wlan0mon
Interface wlan0mon:
ioctl(SIOCGIFINDEX) failed: No such device
Failed initializing wireless card(s): wlan0mon
```